

Course Code : IDT 353

KQLR/RS – 24 / 1175

**Sixth Semester B. Tech. (Electronics and Communication Engineering)
Examination**

BIOLOGY FOR ENGINEERS

Time : 3 Hours]

[Max. Marks : 60

Instructions to Candidates :—

- (1) Draw neat and clean diagram, wherever necessary.
- (2) Assume suitable data wherever necessary.

1.
 - (a) List any five human physiological systems. Elaborate on the human Lymphatic system (part of Immune system) in detail. 5(CO1)
 - (b) Draw well labeled diagram of animal cell. Also elaborate on the functions for at least four cell structures. (i.e mitochondria) in it. 5(CO1)
2.
 - (a) What are the functions of Enzymes in human body ? List any four enzymes found in human body along with their functions. 5(CO2)
 - (b) What are the amino acids and how do they relate to complete protein ? Give the differences between Essential and Non-essential amino acids along with examples. 5(CO2)
3. What is Glycolysis ? Write down all the steps involved in Glycolysis and denote the energy investment and payoff phase. 10(CO3)
4. Discuss the term Genetics. Also discuss the Expression and Transmission of Genetic Information in detail.

OR

Discuss the term Genetics Codes. Also discuss the concept of DNA Cloning in detail. 10(CO4)

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Contd.

5. (a) Explain the origin of electrical activity of the heart with the help of a diagram. Draw a typical ECG waveform and label it. 6(CO5)
- (b) What is the relative importance of the Na^+/K^+ pump in action potential generation and propagation ? 4(CO5)
6. Explain the term Medical Imaging along with examples. Explain in detail about the Magnetic Resonance Imaging (MRI). 10(CO5)

